

Writing Mathematics from the Literature and Reviewed Revisions

A method for reading, revising and retaining useful editorial lessons

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ABSTRACT

A mathematical sentence can be easy to read and still conceal the step that justifies it. Nearby papers offer more useful models than a list of preferred words: they show how the field introduces hypotheses, names objects, connects deductions and allocates space to a proof's difficult transition. We describe a method for studying those choices and testing their use in another manuscript. The examples come from reviewed revisions of eight Erdős-problem papers. They include an infinite construction whose remainder must vanish, an estimate whose required accuracy should be explained before it is proved, and proposed rules that became too strong when detached from their original case. Each lesson retains its source, the old and proposed passages, the integrating decision and a limit on reuse. The resulting guidance extends an existing public writing skill and can be included at an exact version in a later review packet. This is a documented editorial practice; it has not been evaluated as an intervention in reader understanding.

1 Learning to explain a particular argument

Suppose a proof constructs a sequence by repeatedly choosing a digit. A sentence saying that the choice can be repeated explains how to obtain every finite prefix. It may leave the reader to supply the reason why the resulting infinite series has the prescribed sum. Replacing the sentence by smoother prose does not address that difficulty. The explanation needs the invariant for the remainder and the fact that its bound tends to zero.

This example, developed in Section 2.1, suggests where to look for a useful model. Another paper that constructs an infinite sum through controlled finite remainders can show how to separate the choice from the limiting argument. It need not study the same sequence. Its relevance lies in an identifiable part of the proof, and the comparison ends where the arguments differ. The model contributes an expository choice; the new paper must supply its own mathematics.

General advice remains useful. Halmos discusses audience, organization and the selection of examples [1, §§3–4]. Knuth, Larrabee and Roberts discuss the introduction of

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

symbols, the connection between sentences and the reader's need to know why a step is being taken [2, §1]. Gowers examines the placement of examples relative to unfamiliar abstractions [3], while Tao cautions against sacrificing usefulness to excessive optimization of a paper [4]. Such advice identifies questions to ask. A nearby mathematical argument shows how an author has answered them in a particular setting.

The method described here grew out of revisions to short papers and longer research records for eight Erdős problems. Most research and drafting in this project was performed by AI agents. Some agents worked from frozen packets; others had access to the live repository and integrated proposed changes. Their judgements are editorial evidence, with the limitations described below. They are not observations of how an independent mathematical audience read the papers. The examples support an inspectable procedure for revision rather than a claim that the procedure has already been shown to produce better readers or better mathematics.

1.1 Choose sources by the work the prose must do

Before collecting stylistic examples, identify the manuscript's subject, genre and reader. An irrationality note may need a compact route from an assumed rational sum to a positive integer smaller than one. A geometric paper may need to explain which component is being measured and why a map preserves the quantity under discussion. A survey has to distinguish several viewpoints; a short theorem paper may need to develop just one. These differences affect terminology, notation and the amount of preparation a proof needs.

Choose a small group of human-authored papers that addresses these needs. Include a paper close in subject for vocabulary and attribution, and, when useful, one close in proof structure for the explanation of a construction or estimate. A famous paper from a remote field can offer a comparison, but it cannot settle the conventions of the target field. Nor does frequent usage make a term appropriate when its definition differs from the object at hand.

Read the original passage in a named version. An abstract can establish the name used for an object; it seldom reveals the pacing of the proof. For that, read the argument around the sentence, including the statement it proves and the hypotheses it invokes. Record the section, theorem or equation label and page. Preprint versions can have different numbering from a published article, and a source archive may differ from the accompanying PDF. The file's digest identifies the supplied bytes; a public link tells the reader where to seek the work. Neither should silently stand in for the other.

The record should distinguish three acts. A packet may contain a source. Its recipient may declare that particular pages were read. An integrating reviewer may later inspect the quoted passage. These are different facts. Later inspection can verify a locator or expose a poor analogy; it cannot establish what the earlier recipient actually read.

1.2 Read sentences as mathematical relations

A useful reading pass asks what each choice allows the reader to infer. Consider the placement of a hypothesis. A theorem statement should quantify the objects and state

the assumptions needed for its conclusion. Inside the proof, recalling one of those assumptions can explain a step that would otherwise look automatic. A phrase such as “by positivity” helps only if the reader can identify the positive quantity and the inference that uses it.

Connective words deserve the same attention. “Since” introduces a reason; “hence” asserts that the conclusion follows; “provided that” makes a condition visible; “it remains to prove” identifies unfinished work. These words are not interchangeable devices for varying a paragraph. A revision that changes “provided that” to “hence” can change the logical force of the sentence while leaving every displayed formula intact.

Parallel syntax helps compare parallel claims. A subordinate clause can keep a dependency beside the assertion that uses it. There is no preferred ratio of short to long sentences: a short sentence may hide its antecedent, while a longer one can keep a condition attached to its consequence. The point of studying an author’s construction is to understand that relation, then write original prose appropriate to the new argument.

Notation should receive this close reading as well. Ask when a symbol first becomes necessary, what repeated work it saves and which nearby quantities the reader might confuse with it. A new name for a standard object creates an additional translation. Conversely, removing an established term such as “Stieltjes moment sequence” can make a manuscript less connected to its literature [7]. A vocabulary pass should compare definitions and uses, rather than delete unfamiliar words merely because they are technical.

Proof pacing is the allocation of explanation to mathematical difficulty. The reader may need to see the accuracy required of an estimate before its derivation, or the obstruction that motivates a parameter choice. Several lines of routine substitution can be shorter than one indispensable sentence about why a limit exists. The manuscript should give the latter its space. These choices can be examined in actual passages, without imitating an author’s voice or turning a preferred cadence into a rule.

2 Two revisions and the mathematics that permits them

The following passages come from the short-paper revision round dated 30 September 2026 and labelled R6. This label is local to that series of packets; an earlier September series also used round numbers. The public lesson records retain the series, problem and source document so that the two histories cannot be confused. The accepted manuscript passages below are recoverable at repository commit 18cedadedd1, whose full identity and source digests appear in those records [8].

2.1 From repeated choice to an attained infinite sum

The #251 paper contains a construction in which a target x is represented by chosen digits d_j and weights w_j . A finite remainder ρ_j is kept between 0 and a bound F_j . The old passage began:

Given $x \in [0, F_0]$, we choose d_0 so that $x - d_0 w_0 \in [0, F_1]$ and repeat this choice for each successive remainder.

This describes the repeated choice. It leaves the reason for attainment less prominent than the construction. The reviewed addition states:

The covering just proved makes this possible at every stage. The invariant $0 \leq \rho_j \leq F_j$ and $F_j \rightarrow 0$ then give

and continues with the limiting identity. The two sentences distinguish the finite covering argument from the passage to the infinite sum. The record is r6-251-style_rules-04; the full identifier carries the series date.

Crmaric and Kovač's proof of Lemma 4(a), in the first version of their paper on sums of reciprocals, provides a close model [5, pp. 4–5]. They control a finite remainder in an interval, then use convergence to obtain the represented value. This is more specific than an instruction to add intuition. It identifies two jobs in the proof and gives each a sentence.

The analogy has a definite boundary. For the #251 manuscript, the covering property, the invariant and $F_j \rightarrow 0$ must all hold for its own construction. A finite digit example proves none of them. Nor does attainment alone prove the sparsity or arithmetic constraints imposed elsewhere in the paper. The integrating decision preserved those separate obligations and the unresolved status of the original prime-gap problem. The transferable lesson concerns the exposition of an already justified limit, not permission to supply a missing convergence argument by analogy.

2.2 State how accurate the estimate must be

In the #269 argument, a rationality assumption places normalized tails Y_a in the finite set

$$(K^{-1}\mathbb{Z}) \cap (0, 1).$$

Here $K \geq 1$ is the integer denominator in the rationality assumption. An accepted sentence then says that equality of two tails still has to propagate. This matters: finiteness provides repetition, but a repeated value need not determine the subsequent choices. A separate recovery argument is needed before one concludes that blocks repeat.

Hančl and Tijdeman's proof of Theorem 2.1 begins by turning a rationality assumption into an integral scaled remainder [6, p. 373]. That ordering makes the arithmetic object visible before estimating it. The #269 revision uses the same explanatory idea while retaining its additional recovery problem. The cited proof does not establish that extra step.

The next difficulty is quantitative. The maps $G_{53}(t) = (9 + t)/30$ and $G_5(t) = (3 + t)/10$ encode two possible blocks of prime-power jumps. Their images overlap on the initial tail range $[0, 1]$. The earlier paragraph observed the overlap and announced that the range would be narrowed. With the lower bound $3/10$ already established, the revised passage specifies the required accuracy first:

With that lower endpoint, separating these two images requires an upper endpoint U satisfying

$$G_{53}(U) < G_5(3/10), \quad \frac{9 + U}{30} < \frac{33}{100}, \quad U < \frac{9}{10}.$$

This specifies the improvement needed from the arithmetic sequence.

The case is `r6-269-style_rules-03`. The inequality now explains what the following estimate is trying to achieve. It also makes a checkable distinction: this is the sufficient threshold used by the argument, not a claim that no other method could work with a different bound.

The return associated this choice with an item in Knuth’s notes about the author’s dialogue with the reader. That is a broad compositional analogy, not a source for the displayed threshold. The algebraic separation condition is the precise local reason for the new sentence. Keeping the two explanations separate prevents the general writing reference from appearing to support mathematics it does not discuss.

2.3 An example must reveal the relevant difference

The same manuscript distinguishes counting smooth integers from counting their distinct heights. Its revision groups the seven smooth integers in $[16, 32)$ by height:

Smooth integers	Height	Contribution
16, 18, 20, 24	720	4/720
25	3600	1/3600
27, 30	10800	2/10800

The multiplicities 4 and 2 are the information lost by replacing repeated heights with distinct ones. This table explains a choice of counting convention; its purpose is not to decorate the definition. The associated record, `r6-269-style_rules-04`, also preserves an endpoint warning: the distinct-height argument uses $(16, 32]$, whereas this table uses $[16, 32)$. The writer must explain that change rather than identify the two intervals silently.

Halmos’s discussion of concrete cases and Gowers’s discussion of examples suggest asking whether an instance can reveal the unfamiliar operation before its notation accumulates [1, §4][3]. The usefulness of this particular instance is more narrowly demonstrated: one can see the lost multiplicity in the table. It does not establish the infinite-series theorem or show that every reader would prefer this placement.

3 What the integrating reviewer must decide

A proposed revision has several possible fates. It can be accepted verbatim, accepted after a mathematical or expository repair, rejected, or left pending a named check. Its lesson may already be expressed in the writing guidance. These decisions should not be collapsed into a single “accepted” label. The revision of a paragraph and the adoption of a general rule are different acts, as are admission of a returned archive and acceptance of the mathematics inside it.

The reviewer begins with the current statement and proof. For a substantive change, the record retains the old passage, the proposed passage, the source of the proposal and the decision actually made in the canonical manuscript. A digest identifies the source bytes; a commit, path and label identify the accepted passage. An exact text match, after

whitespace normalization, can help locate it. Such a match does not establish equivalence of two theorem statements or prove that a general recommendation is sound.

3.1 Replace a local term without losing its definition

The next short-paper round, R7, provides a direct example of terminology review. In the #249 record, a local growth adjective was replaced by the condition it denoted:

Before: tempered integer carry orbit.

Selected revision: integer carry sequence u with $u(N)/2^N \rightarrow 0$.

The gain is specific. A reader no longer has to recover a private definition to know the required rate. It would be incorrect to replace this condition by the more familiar phrase “subexponential growth”, which says something stronger. Other revisions retained standard terms such as upper Banach density because those terms denote the properties the paper actually uses. The rule is to compare meanings, not to replace every unusual expression.

Removing a shorthand can also remove the visible source of its assumptions. In #243, “positive exact state” bundled recurrence and positivity conditions. A proposed replacement referred to the same assumptions after that definition had disappeared. The integrating reviewer instead stated the recurrences for the first implication, then explicitly added natural-number domains and $a_n > 1$, $C_n > 0$ for the eventual consequence. In a second passage, the reviewer removed “earlier” from “earlier decreases”: the original qualification did not restrict the time at which decreases could occur.

The R7 lesson records distinguish these reviewed selections from the donor’s proposal and from later rendering or formal correspondence checks. These cases show why terminology, grammar and quantification belong in the same review. A plainer noun is useful only if the sentence still makes the same claim.

3.2 Rejecting an attractive universal rule

Earlier reviews of the #243 manuscript illustrate why historical advice needs its original setting. One round asked for a cubic irrationality example near the beginning. A subsequent round explicitly retained the live bounded-negative result as the paper’s lead and rejected an invariant rule that an irrationality result must always come first. Later editorial routes treated page targets as subordinate to the proof’s dependencies.

The retained lesson is to choose a coherent principal contribution and explain its relation to the named problem. It does not prescribe a theorem category or a fixed page on which every proof must finish. A conditional reduction can be the useful result; an unconditional statement can be too weak to carry the paper. The choice requires comparing what the results actually say. The history record preserves the successive recommendations instead of combining them into contradictory commands.

This distinction also protects the longer research record. Shortening a paper can be appropriate when the subordinate details have a complete, checked destination. Regenerating the long record from the shortened version would lose the very argument that made the shorter presentation possible. The two documents must be compared in both

directions: a stronger hypothesis, a repaired endpoint or a newly explained limiting step may appear first in either one.

3.3 A better account can require a repaired argument

An earlier #251 review identified a defect in the manuscript's condensed account of a sparse construction: a buffer quantity was used without being assigned a sequence coordinate and included in the value, size and capacity accounting. The original returned memorandum did include those coordinates and their contribution to the weighted sum. The accepted ordinary repair assigned the relevant coordinates and performed the accounting across all indices. Calling the original paragraph unclear would have understated the issue. The condensed proof needed repair before the account of it could be trusted. Here the recoverable original prevented a loss during shortening or integration from being attributed to the original construction.

That decision must be distinguished from a later sharp companion construction which remained unreviewed at its recorded disposition. The presence of both in a returned bundle does not give them the same status. A usable history names the particular construction, the defect and the accepted replacement; it does not summarize the whole problem's sequence of returns as progress through increasingly strong theorems.

Other cases expose similar changes of meaning in small amounts of prose. In #1049, language suggesting that a degree bound was inevitable was repaired to state a sufficient inequality and preserve the qualification contributed by a remaining factor. In #1041, the revision retained the fixed-polynomial scope instead of turning it into a freely varying family. For mixed irrationality criteria in #257, independently chosen witnesses cannot replace the common witness required by the argument. These are local mathematical checks, even when the proposed edit was presented as compression or style.

An ordinary proof and its formal support also require separate decisions. An accepted ordinary argument need not already have a complete Lean formalization. Conversely, a checked lemma nearby does not check every sentence in a new proof. Review should identify the exact statement supported by each kind of evidence, leaving named inputs and pending comparisons visible. The publication system's broader evidence responsibilities are described separately [11]; the writing method uses those distinctions rather than creating another proof registry.

A late-September integration supplies a concrete limit on deleting operational language. A prose pass had removed remarks that identified named mathematical inputs beside the results depending on them. The integrating agent restored those remarks. Routine details about running a check could recede, but the reader still needed to know which assertion depended on an input. Calling both kinds of sentence workflow commentary had erased a substantive qualification.

3.4 Correct the source without rewriting its history

The R6 #1049 return attached a reader-motivation specimen from Knuth's notes to §1, item 18. Inspection of the original places the passage in item 12, on printed page 3 (PDF page 5). Item 18 concerns a different matter. The public lesson record retains both the reported locator and the correction.

This small example explains why a plausible quotation and a plausible page number are insufficient. A phrase search can find the table of contents, another occurrence of a common sentence, or text from an adjacent PDF column. The reviewer must inspect the intended passage in context. A corrected locator does not invalidate an otherwise useful revision, but the revision's success does not excuse a false citation.

The same care applies to the role of the source. A paper may supply a theorem, historical attribution, established terminology or an example of composition. Those uses call for different claims. A paragraph about proof pacing does not validate the proof being paced. A literature specimen establishes that an author made a particular choice, not that this choice is universally preferable.

4 Keeping a useful practice small

The public implementation extends the repository's existing mathematical writing skill [9]. It adds a literature-reading pass before a substantial rewrite and places the detailed cases in a linked guide and lesson records. The two-page *Writing a Good Mathematical Paper* [10] distils the instructions for use without this repository. The present companion explains their evidence and limits. The skill remains a set of working instructions; the records make its examples and exceptions recoverable. A writer need not read every past return before deciding how to explain a particular proof.

The compact rule set asks writers to select a coherent result, use examples for a stated purpose, explain difficult inferences, name the actual mathematical objects and preserve statement meaning. It also covers the relation between short and long papers, evidence, citation roles, informative failures, rendering and honest reporting of review. These topics are not scores to optimize. Each rule states a limit that matters to its use: an example does not prove an infinite result; a shorter sentence need not preserve quantifiers; a rendered PDF need not explain its mathematics well.

4.1 How the practice developed

The recorded process began with reviewers proposing changes to particular papers: move the result forward, explain a construction, shorten a repetitive section or repair an unsupported implication. Successive returns did not form a single consistent programme. In the earlier #243 reviews, a cubic result and a bounded negative result were each proposed as the lead at different stages. Comparing these requests with the manuscripts yielded a narrower instruction: choose the result that gives the current paper its most coherent argument. The historical request remains evidence about that draft, not a permanent ranking of result types.

Each returned manuscript then met a separate integration decision. In the later second round, only one of nine candidates was accepted, although the returns had passed transport checks. In the third round, integration included restoring seventeen named-input remarks removed during compression. These events exposed different questions: did the proposed files arrive intact, did the revision preserve the mathematics and credit, and did it explain the argument? A pass on the first question could not answer the others. Comparing short papers with their longer records also recovered useful material, as the #251 buffer-coordinate case illustrates.

The recent rounds supplied more explicit literary models, before-and-after passages and proposed general lessons. We compared these with the selected native source, reopened the cited passages and retained repairs alongside acceptances. The finite-remainder explanation survived this review; a claimed Knuth locator needed correction; a broad author–reader analogy did not by itself justify a specific rule about estimates. The resulting instructions therefore combine mathematical checks with reading practice, while the records preserve the disagreements and limits that compression would hide.

This short/long pair is the next distillation of that sequence. The compact guide gives the writer actions; this companion explains why those actions are useful and when they are insufficient. A later packet can carry both at a fixed version, solicit another local revision and repeat the review. The systems paper [11] describes the repository mechanisms supporting the process. Here the object of review is the editorial decision itself.

4.2 From one case to a bounded amendment

Most accepted changes should become examples of existing guidance. Add a rule only when the case exposes a reusable distinction that the present instruction misses. State the circumstance in which it applies, the action to take, the evidence motivating it and a case in which it would fail. A page-one request can become a reminder to remove unnecessary delay without becoming a page-one quota. A failed proof can become an explanation of a missing hypothesis without becoming a claim that every alternative approach is impossible.

The public records separate recent proposals, historical requests, source identities and the manuscript decisions used to assess them. They retain unmatched and pending cases explicitly. A return may describe an edit in summary rather than supply the exact accepted words; this should be recorded as a correspondence still requiring inspection, not counted as either a rejected edit or a verbatim acceptance. Source passages are reviewed for their stated writing use. Inclusion in the source list does not assert a complete mathematical review of every cited paper.

The record of a failure should be equally specific. Preserve the attempted implication, its assumptions, the witness to failure and the information missing from the argument. Failure of one estimate does not refute the theorem. Failure to run a check says that the check is unrun. An unavailable source says something about the review’s evidence, not about the truth of the source’s claim. These distinctions prevent a later writer from giving a historical obstacle more force than it had.

4.3 What the next packet must contain

A frozen review packet should include the exact skill and guide it asks the recipient to follow, the compact paper, the relevant lessons, the source identities and this companion paper. A manifest records their paths and byte digests. The packet must also carry the primary originals needed for its selected specimens, with versions and reading-status declarations kept separate. A title in a bibliography or a list of annex files is not a substitute for a source that the recipient can actually inspect.

The recipient can then compare those papers with the supplied manuscript: how does each introduce its object, make a hypothesis visible, motivate an estimate or credit an input? The return should identify the passage read, the feature adapted, the proposed local change and its limit. The repository agent reviews the proposal against the current mathematics, integrates an accepted change and amends the guidance only if the evidence warrants it. The next packet can consume that resulting version. This is a sequence of explicit document revisions, not training of model weights.

Freezing a version matters because guidance can change during a review. An earlier packet cannot have applied a later correction to Knuth's locator, and a later integration should not attribute that correction to the original return. Keeping both versions makes the difference inspectable. The same principle applies to mathematical status: a later accepted ordinary proof does not turn an earlier unassessed proposal into an already checked result.

4.4 Read the page that will be read

The final pass uses the repository's native sources and rendering tools. After changing a figure, inspect it at the size in which it appears. Follow a reference to the intended theorem or long-form argument, rather than merely checking that a target file exists. After reflow, inspect the affected pages and their neighbours: a useful explanation can be separated from its figure, or a small heading can be stranded above a page break. Bibliographic labels, captions, interval endpoints and cross-document links belong to this review.

The checks answer limited questions. A successful build establishes that the source rendered under the recorded toolchain. A passage comparison helps establish what changed. A source inspection supports a citation's locator and role. Neither those checks nor a lower rate of phrases flagged by a style detector establishes improved comprehension. A reported style comparison must identify the versions, prose denominator and genre, and retain regressions alongside gains. A claim of improved comprehension would require evidence from readers performing a specified task on identified versions, with the comparison and its limitations recorded.

The method leaves that empirical question open. Its present contribution is to make editorial reasoning recoverable: a nearby paper offered a model, a local argument licensed a revision, and a reviewer accepted, repaired or rejected it for stated reasons. A different proof may warrant a different choice.

Availability and authorship

The writing skill, guide, source records, lesson history and this paper's LaTeX source are maintained in the public repository [8, 9]. The paper studies recorded revisions and limited inspections of cited source passages. It reports no new mathematical result and no controlled study of readers. Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

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